

Method 1 (Ground-Distance) Slope Estimation

Cross-application to the Method-2 dataset · per-segment slope analysis · Robinson plan

Drone terrain slope from VGGT reconstructions

What was asked, and the short answer

- The request

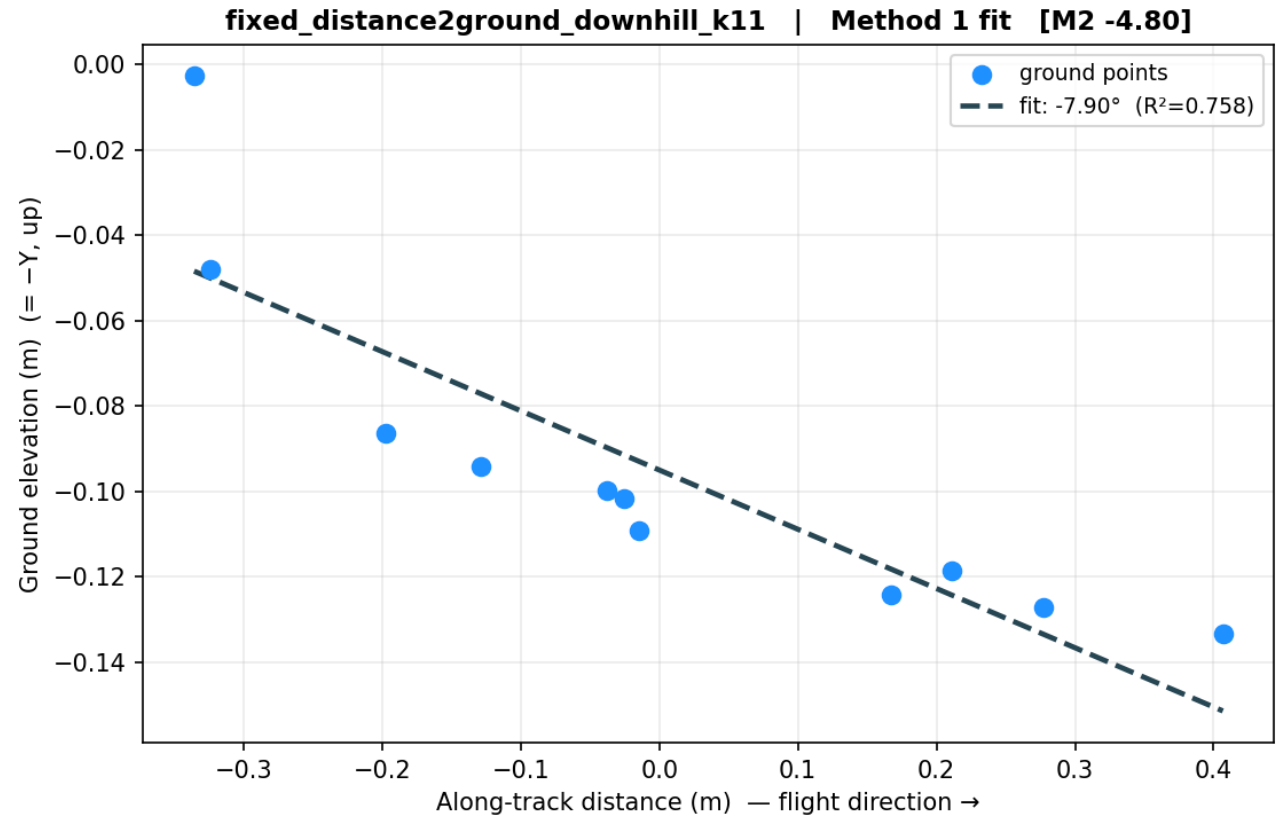
- Apply Method 1 to the dataset used by Method 2
- Method 1 is more general — it doesn't assume fixed height above ground
- Process the two Robinson datasets with Method 1
- Report slope with a sign (+ uphill / – downhill); use all images
- Per-segment slope plots; same plots for the previous dataset
- Collect everything into a slide file

- Short answer

- Method 1 transfers to the Method-2 dataset and works well
- Uphill: $+3.7^\circ$ (Method 2: $+3.4^\circ$) — close agreement, $R^2=0.99$
- Downhill: -7.9° (Method 2: -4.8°) — right direction, steeper
- Method 1 needs the reconstruction to capture vertical relief (terrain-following flights) — level flights rebuild flat
- Robinson uphill: $+16.3^\circ$ uphill (GPS-verified); downhill: -5.65° downhill
- Sign finding: Method 1's up/down sign rides on VGGT's vertical axis, which is NOT gravity-locked — we now anchor the sign to GPS

How Method 1 works (ground-distance method)

- Assume the GLB's up-axis (+Y) points along gravity
- For each camera, look straight down and take the road surface beneath it (point cloud)
- Collect those ground points along the whole flight
- Fit a line: ground elevation vs. distance along the path
- The tilt of that line is the slope
- It reads the terrain directly — so it does NOT need the drone to keep a fixed height (that is Method 2's assumption)



The actual fit: ground elevation vs. along-track distance — its tilt is the slope.

Method 1 vs. Method 2 — what each assumes

	Method 1 — ground-distance	Method 2 — camera-altitude
What it measures	Terrain points beneath the camera	The camera's own up/down motion
Key assumption	Up-axis = gravity	Drone keeps a fixed height above ground
Generality	Works for any flight (if relief is reconstructed)	Only valid for terrain-following flights
Fails when	Reconstruction comes back flat (no vertical parallax)	Drone altitude $\neq$ terrain (e.g. level flight)

- Shengqing's point is right: Method 1 is the more general of the two — it never relies on the fixed-height assumption.

Sign convention (so + / – is unambiguous)

- All images are used (1, 2, 3, ...) — no frames dropped.
- The sign is anchored to image order — the direction the drone actually flew:
 - + = terrain goes UPHILL as the flight progresses
 - – = terrain goes DOWNHILL as the flight progresses
- Each dataset reports one overall slope (a single line fit over all images) plus a per-segment breakdown between consecutive images.
- R^2 is shown alongside: high R^2 = the terrain profile is cleanly reconstructed; near-zero R^2 = no usable slope signal.

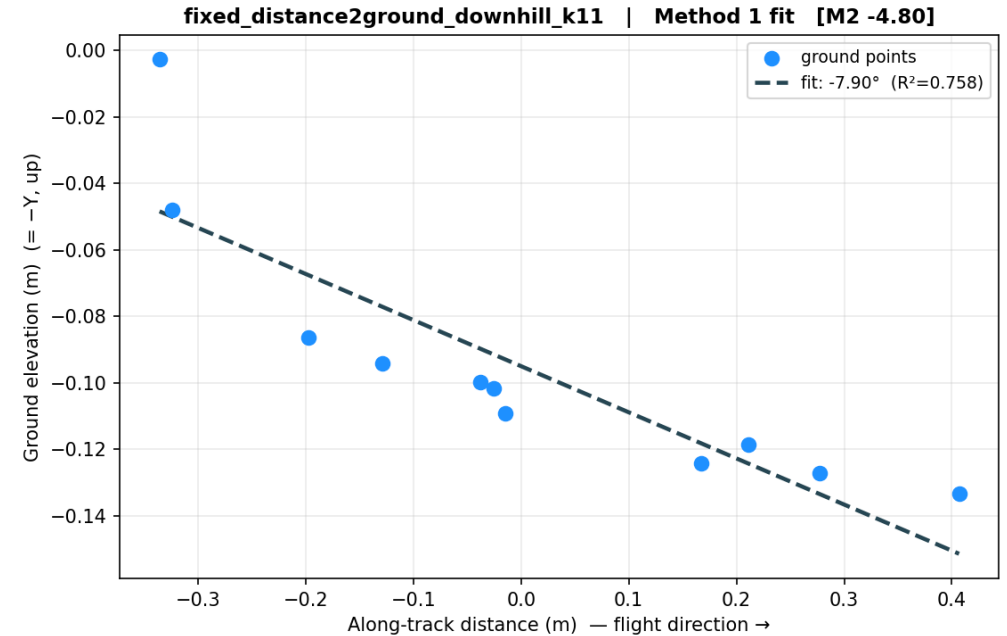
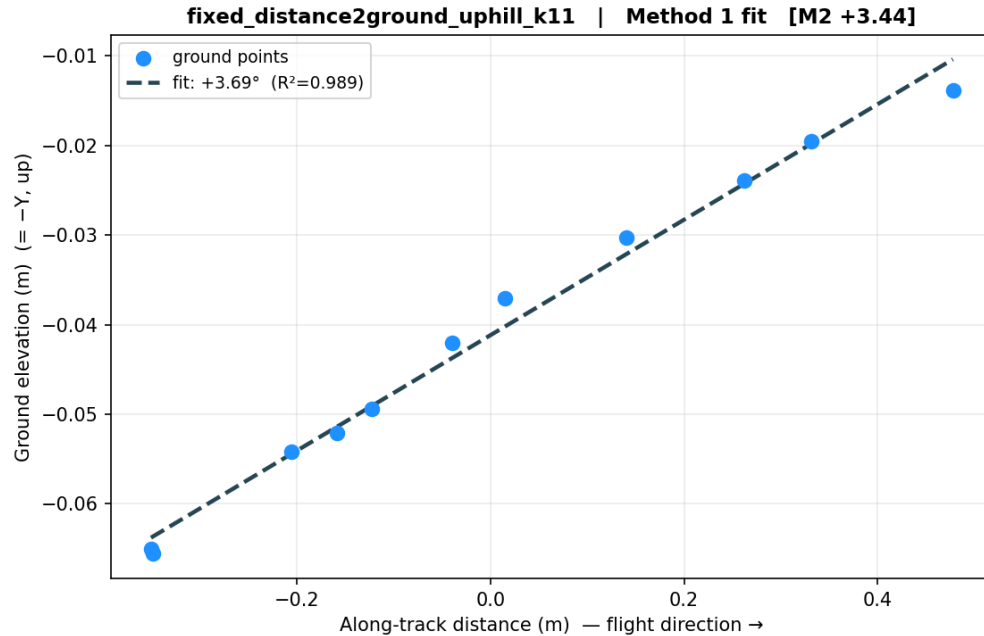
Results — all datasets (Method 1, all images)

Dataset	Method 1	R ²	Method 2	Verdict
Happy Hollow (original)	-3.92°	0.994	—	clean signal
fixed_dist2ground — uphill	+3.69°	0.989	+3.44°	clean signal
fixed_dist2ground — downhill	-7.90°	0.758	-4.80°	clean signal
fixed_altitude — uphill	+0.25°	0.004	-0.12°	no signal (flat rebuild)
fixed_altitude — downhill	-0.11°	0.003	-0.11°	no signal (flat rebuild)

Method 2 numbers shown only for the fixed_dist2ground datasets (where Method 2 is valid).

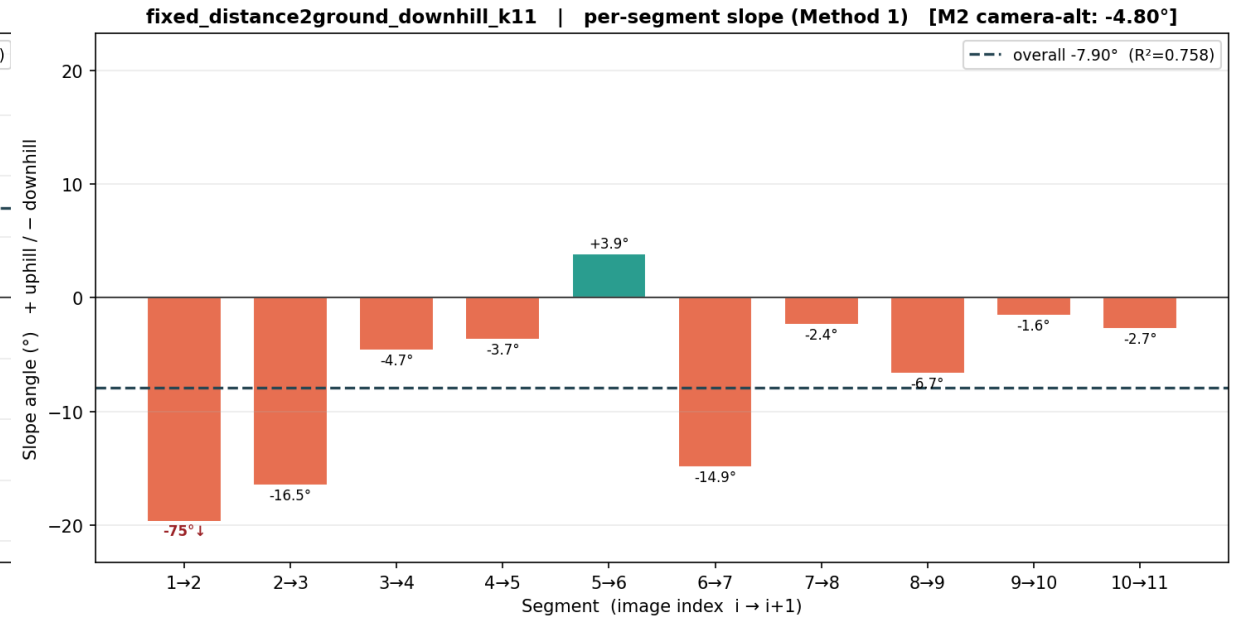
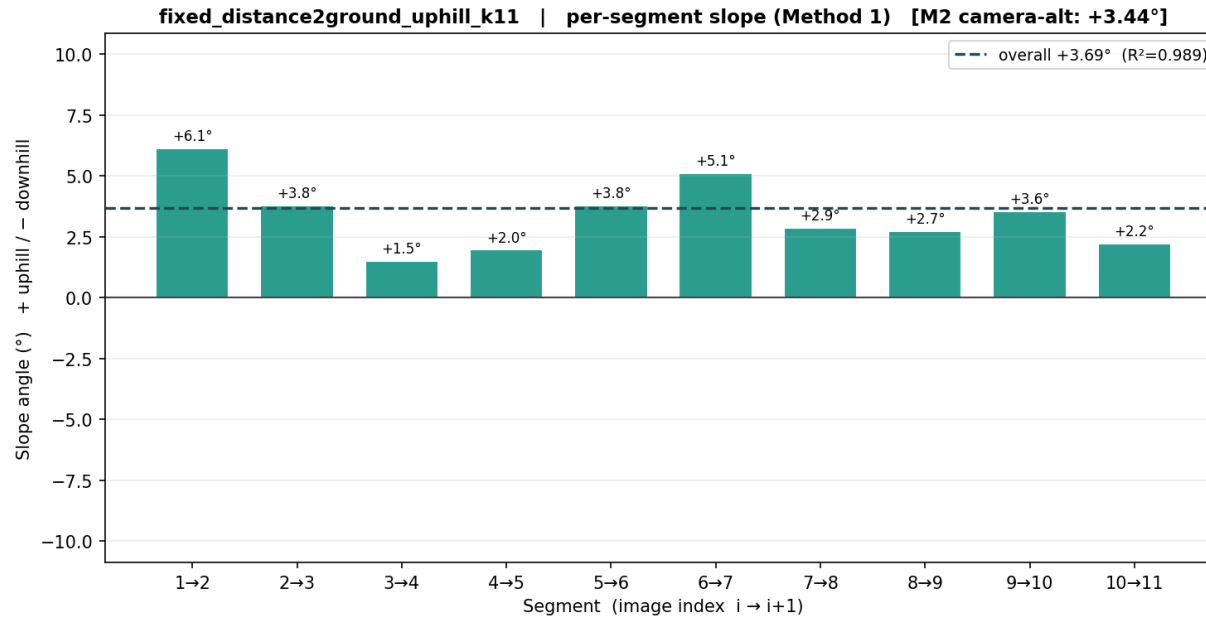
Method 1 on the Method-2 dataset (the main request)

fixed_distance2ground — terrain-following flight



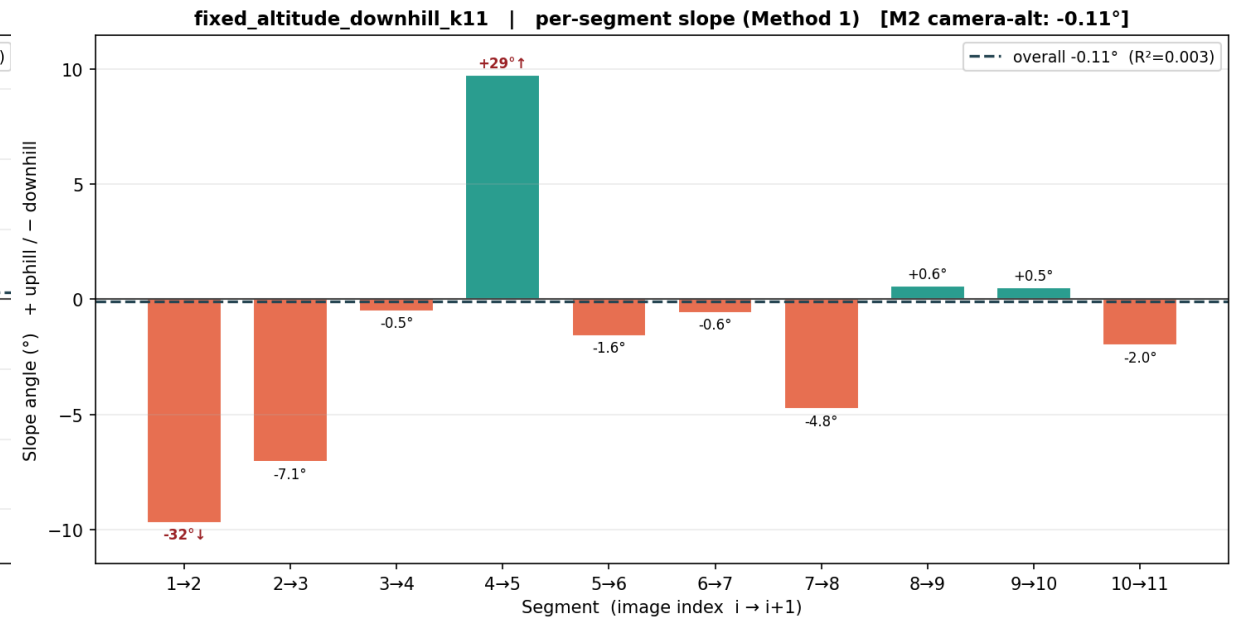
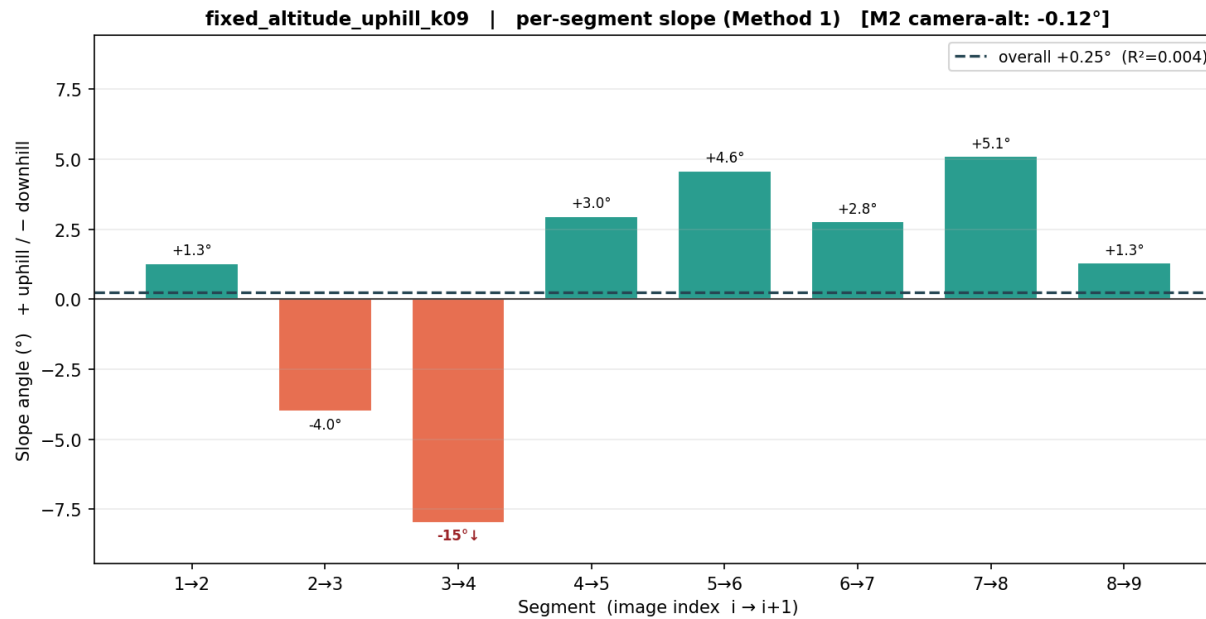
- Uphill +3.69° ($R^2=0.99$) vs Method 2 +3.44° → excellent agreement.
- Downhill -7.90° ($R^2=0.76$) vs Method 2 -4.80° → correct direction, magnitude steeper.
- Takeaway: Method 1 transfers cleanly here because this flight follows the terrain, giving VGGT vertical parallax to reconstruct the slope.

Per-segment slopes — Method-2 dataset (fixed_distance2ground)



x = image-index segment, y = signed slope. Single near-vertical bars (closely-spaced cameras) are capped & flagged.

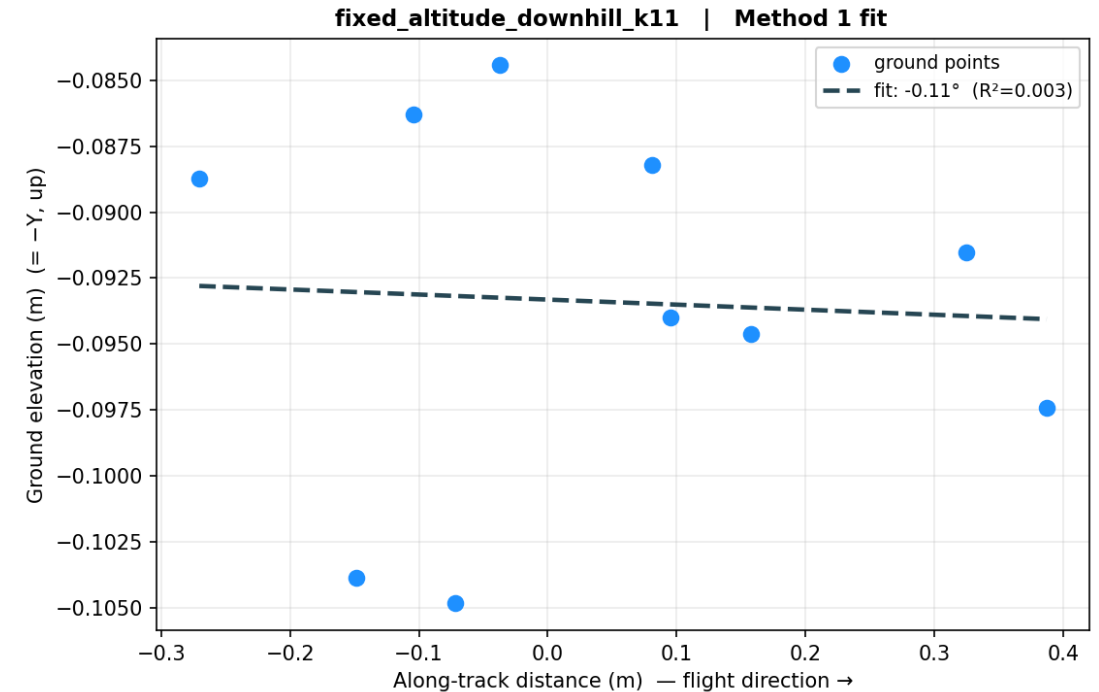
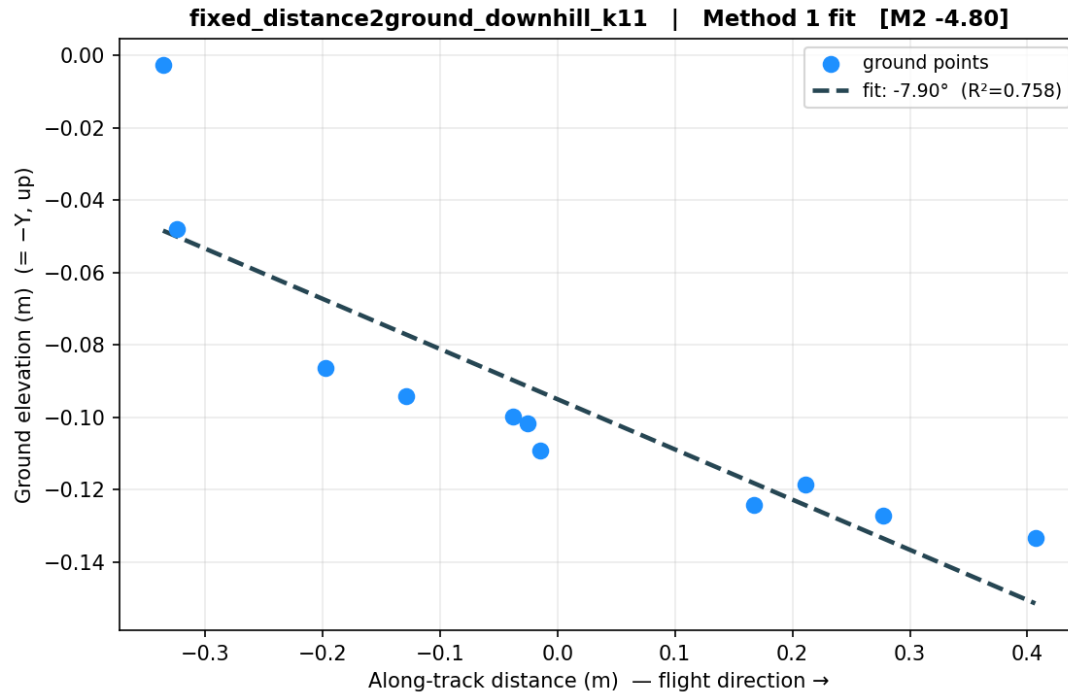
Previous dataset — Method 1 on fixed_altitude (level flight)



- Both come out near 0° with $R^2 \approx 0.00$ — the per-segment slopes are just noise (alternating up/down).
- This is NOT terrain being flat: it is the SAME hill as the fixed_distance2ground flights (matching GPS track).

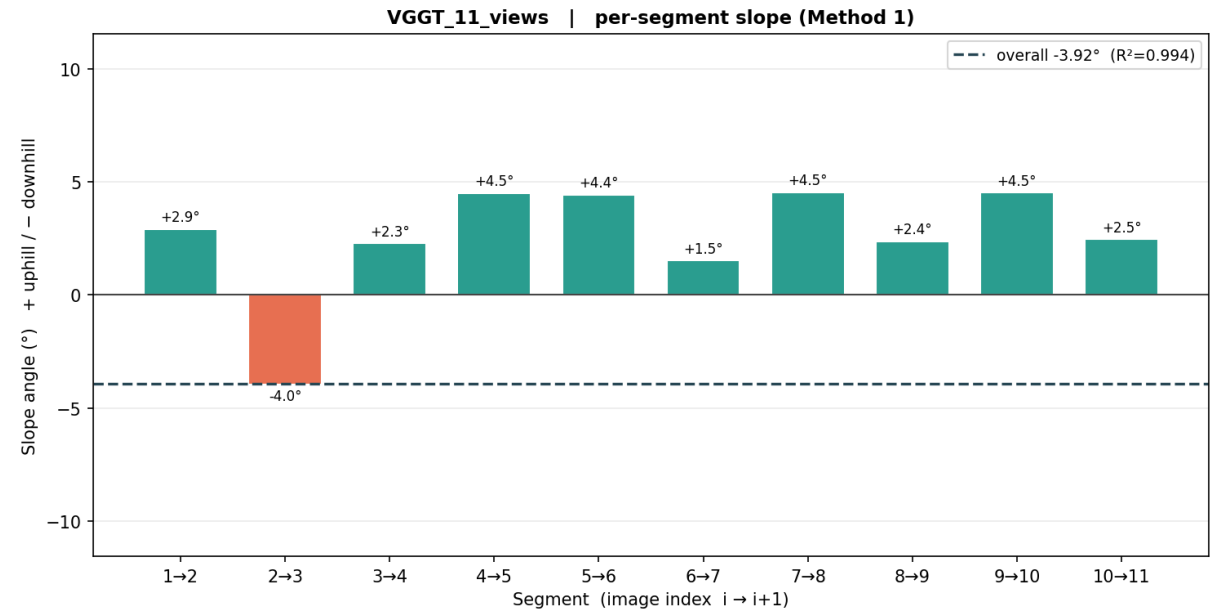
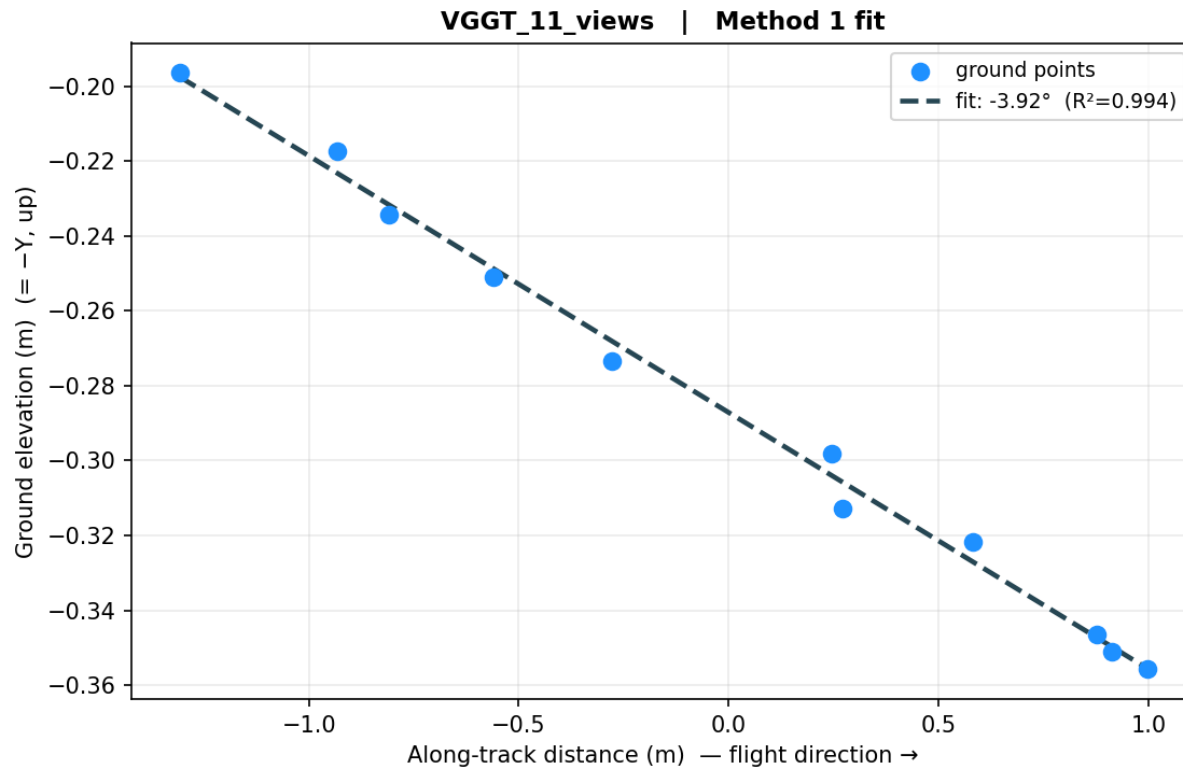
Why: the flight geometry decides whether Method 1 has a signal

Same terrain, same GPS path, two flight styles



- Left (terrain-following): camera moves up/down with the hill → VGGT reconstructs the relief → clean -7.9° fit.
- Right (level / fixed-altitude): camera moves horizontally only → little vertical parallax → ground rebuilds flat → no slope to fit.
- So 'up-axis = gravity' is necessary but not sufficient — the reconstruction must also preserve vertical relief.

Reference: original Happy Hollow reconstruction (Method 1)

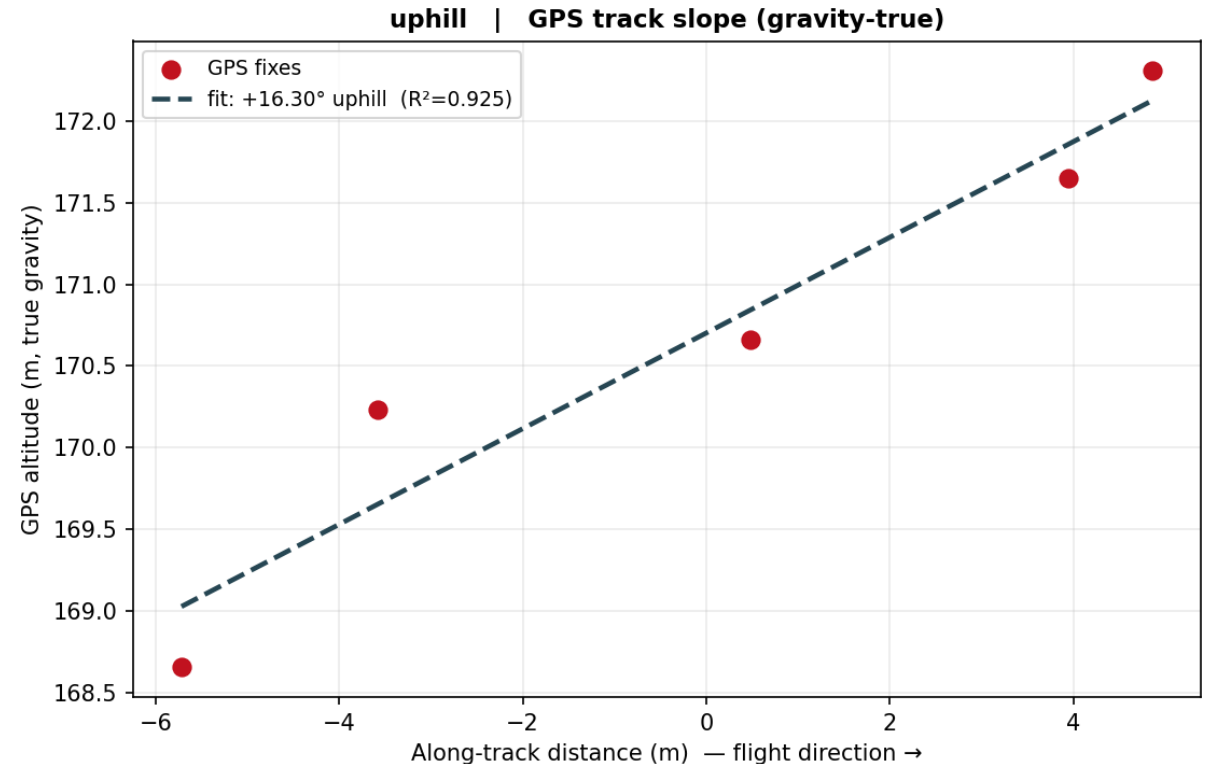


Original 11-view reconstruction: -3.92° downhill, $R^2=0.99$ — Method 1's clean reference case.

Fixing the up/down sign — anchor to GPS, not VGGT

VGGT has no gravity sensor, so its vertical axis (and the slope sign) is not reliable

- Method 1 reads slope in VGGT's coordinate frame, assuming +Y = gravity.
- VGGT never measures gravity — it picks an arbitrary frame — so the up/down SIGN can come out backwards.
- Proof on Robinson uphill: the photos carry GPS, and the drone climbed 168.7 → 172.3 m (clearly uphill).
 - GPS: +16.3° uphill ($R^2=0.93$)
 - Method 1: -2.6° downhill ($R^2=0.29$) ✗ wrong sign & noisy
- Fix: when photos carry GPS, take the sign (and a metric slope) from the GPS track. Otherwise fall back to the known flight label.



Robinson uphill: GPS altitude vs. along-track distance — the gravity-true profile.

Robinson datasets — results

all images, signed (+ uphill / – downhill)

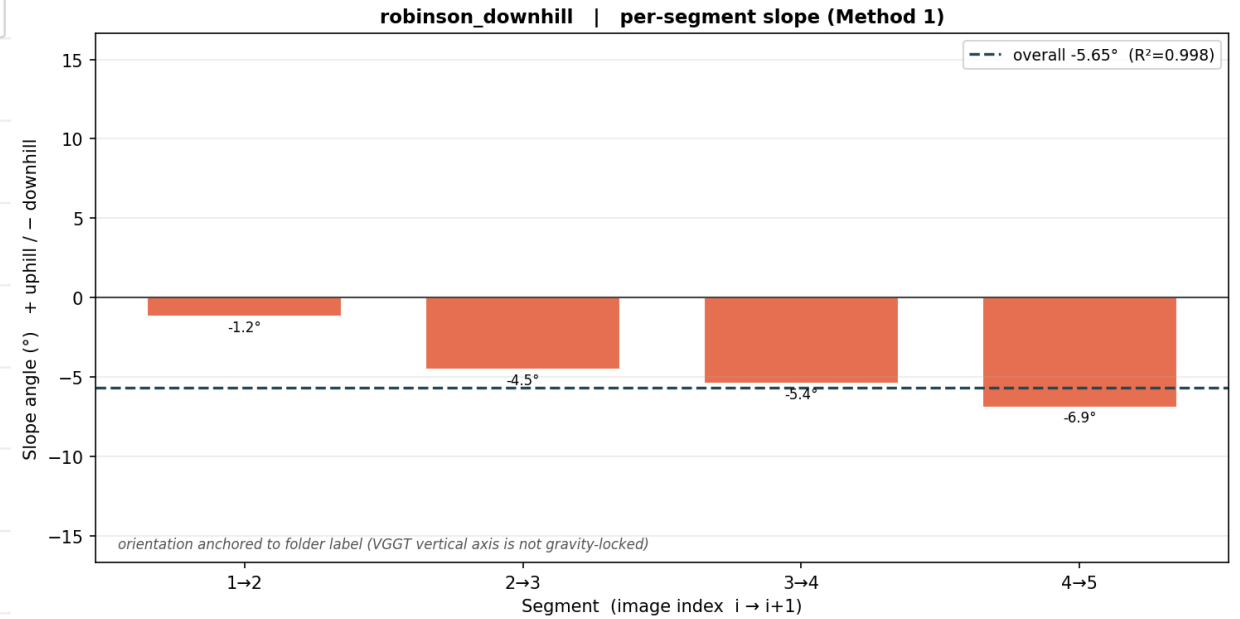
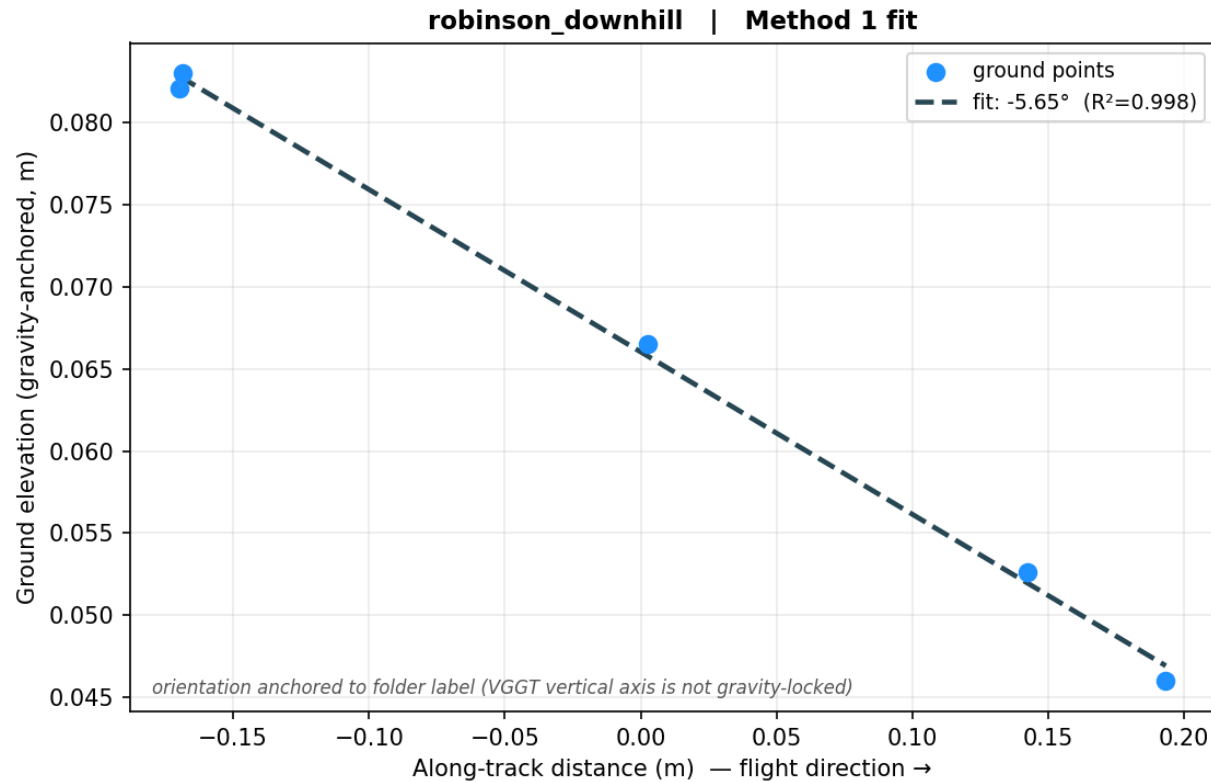
Dataset	Imgs	Method 1 (VGGT)	GPS track	FINAL	Sign from
downhill	5	+5.65° ($R^2=1.00$)	—	-5.65° downhill	folder label
uphill	5	-2.61° ($R^2=0.29$)	+16.30° ($R^2=0.93$)	+16.30° uphill	GPS

- uphill — iPhone 14 with GPS: drone climbed; FINAL +16.3° uphill from the GPS track (Method 1's ground-fit was too noisy here, $R^2=0.29$).
- downhill — DJI, GPS stripped from the files: Method 1's ground-fit is clean ($R^2\approx 1.00$), so we keep its 5.65° magnitude and take the sign from the folder label → -5.65° downhill.
- To verify downhill independently, we'd need its flight log / original GPS — flagged for follow-up.

FINAL sign source: GPS where available, otherwise the known flight label. Magnitude: GPS when Method 1's R^2 is low, else Method 1.

Robinson — downhill

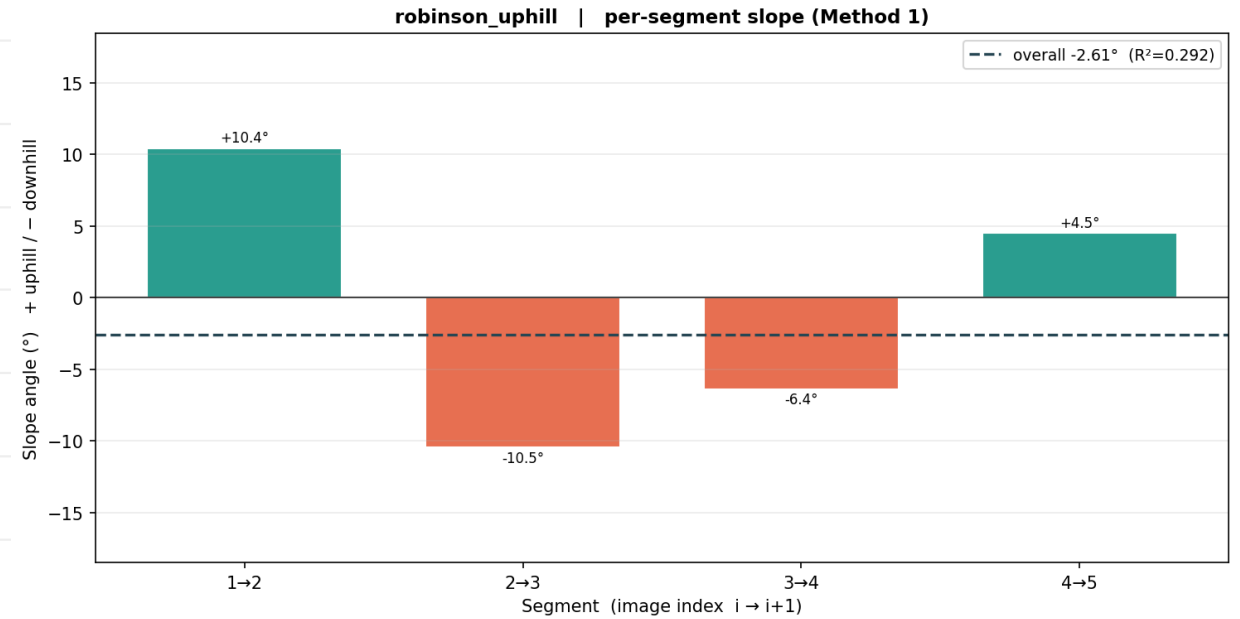
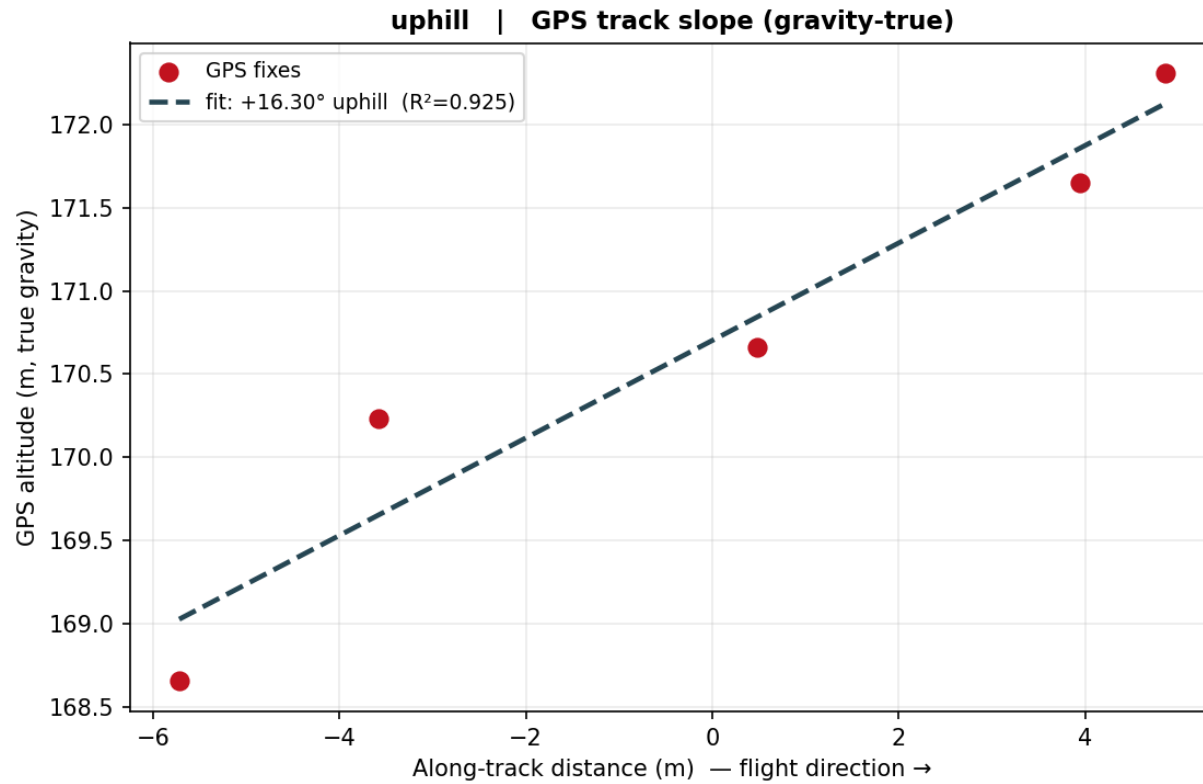
FINAL -5.65° downhill [Method 1 magnitude, folder-label sign] · 5 images



Left: Method 1 ground-elevation fit. Right: Method 1 per-segment slope by image index.

Robinson — uphill

FINAL +16.30° uphill [GPS track] · 5 images



Left: GPS track (gravity-true). Right: Method 1 per-segment slope by image index.

Observations & recommendations

- Method 1 estimates slope MAGNITUDE well when R^2 is high — but its up/down SIGN is unreliable, because VGGT's vertical axis is not gravity-locked.
- Always anchor the sign to a gravity-true reference: GPS altitude is ideal (we proved it flips the wrong Robinson-uphill sign to the correct +16.3°).
- Capture GPS/flight-log with every dataset — without it, the sign can only be taken from the known flight label, which we can't independently verify (e.g. Robinson downhill).
- Reliability tracks R^2 : trust the magnitude when R^2 is high; treat near-zero R^2 as 'no signal', not 'flat ground'.
- Per-segment slopes are noisy and can spike — read them as a profile, not point values; use all images for the overall fit.
- Prefer flights with vertical parallax (terrain-following / oblique) over level passes.

Files & how to reproduce

- `method1_slope.py` — Method 1 (all-frames, signed, image-order, adaptive ground-search radius) + plot helpers
- `gps_anchor.py` — reads EXIF GPS, computes the gravity-true track slope; the sign anchor
- `run_robinson.py` — A100: reconstruct with VGGT (or `--from-glb` to skip GPU), then Method 1 + GPS anchor
- `run_method1_existing.py` — Method 1 on the earlier GLBs → results/ plots + summary.json
- `build_slides.py` — regenerates this deck from results/
- Re-run order: (A100) `run_robinson.py [--from-glb]` → `build_slides.py`